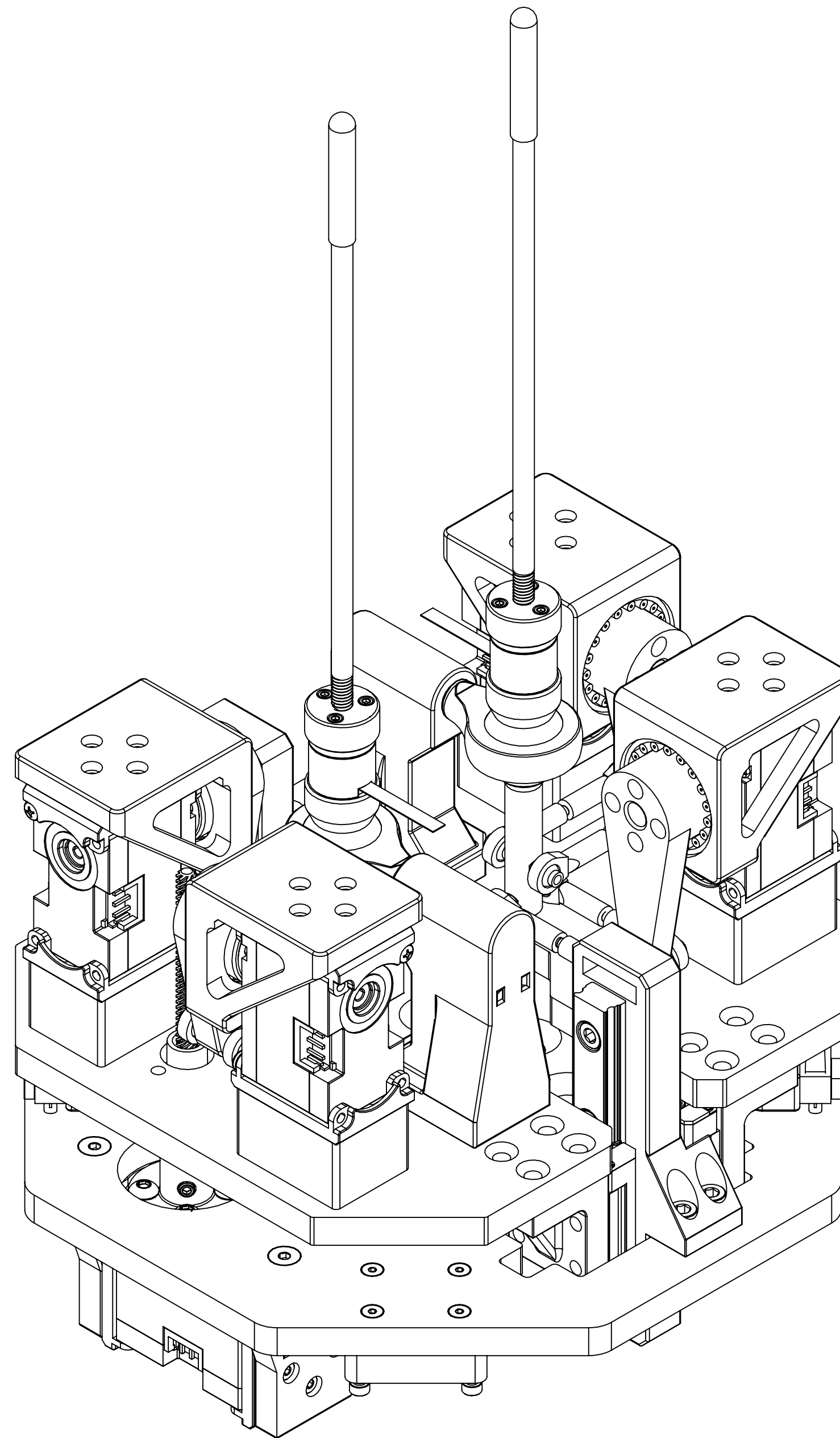


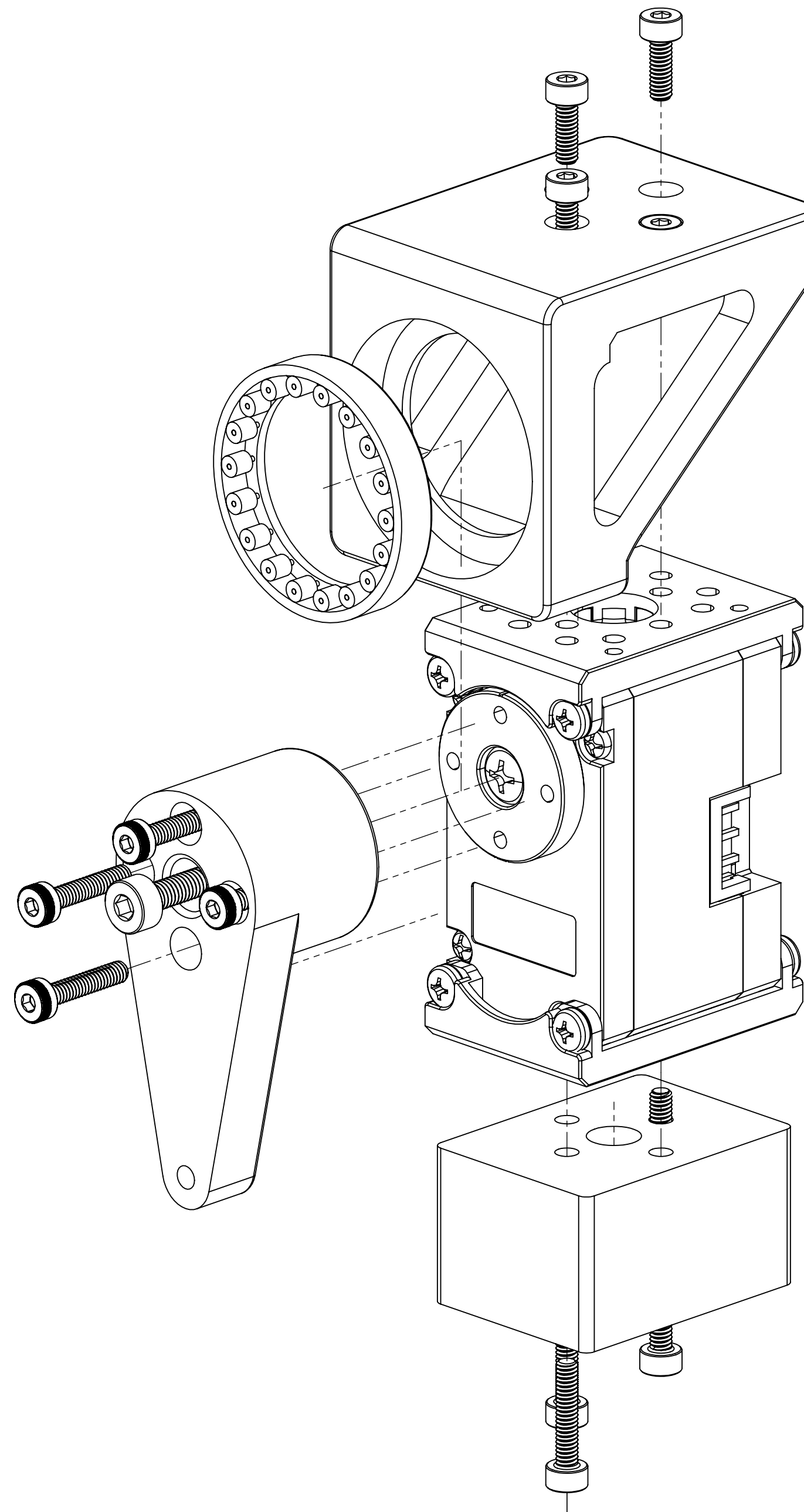
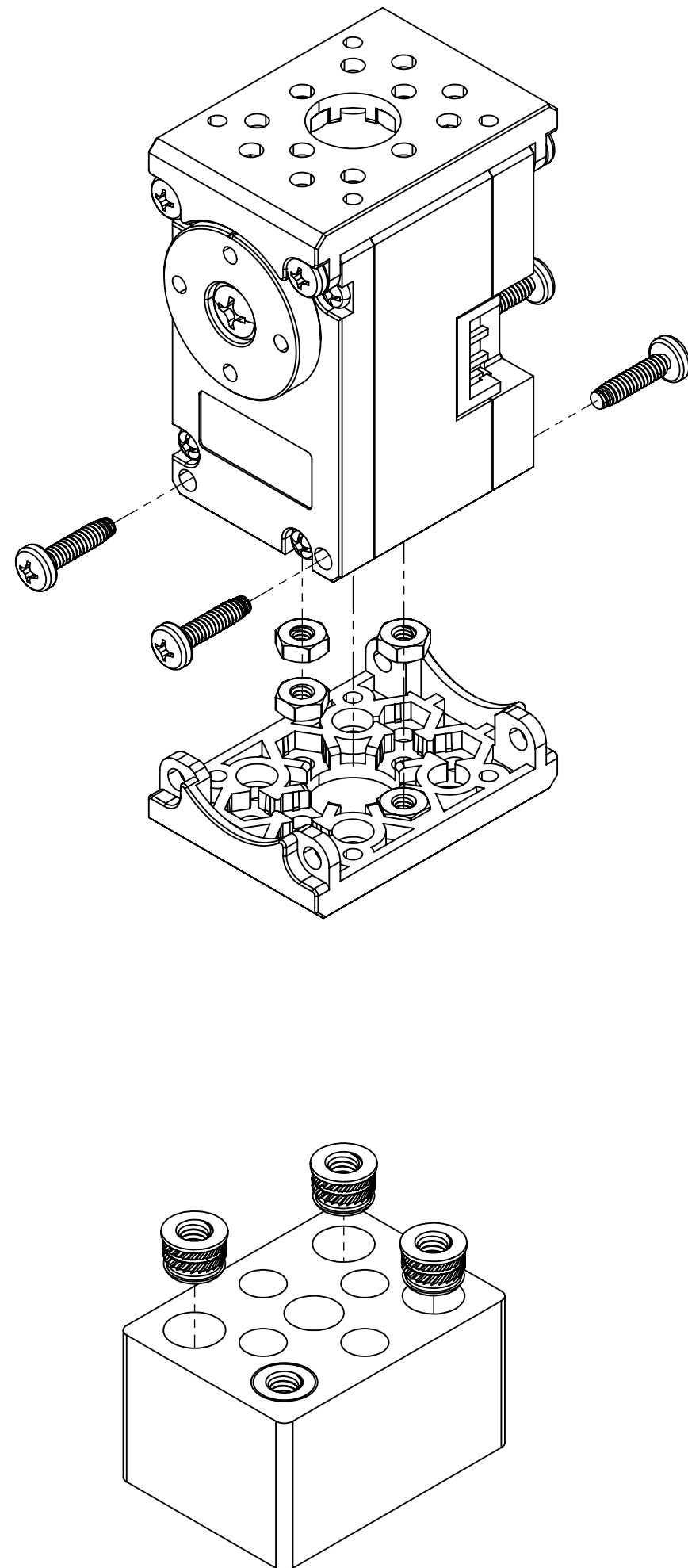
Complete HASHI Assembly



Prepare each Dynamixel by pressing M2.5 nuts into the central four locations in the frames.

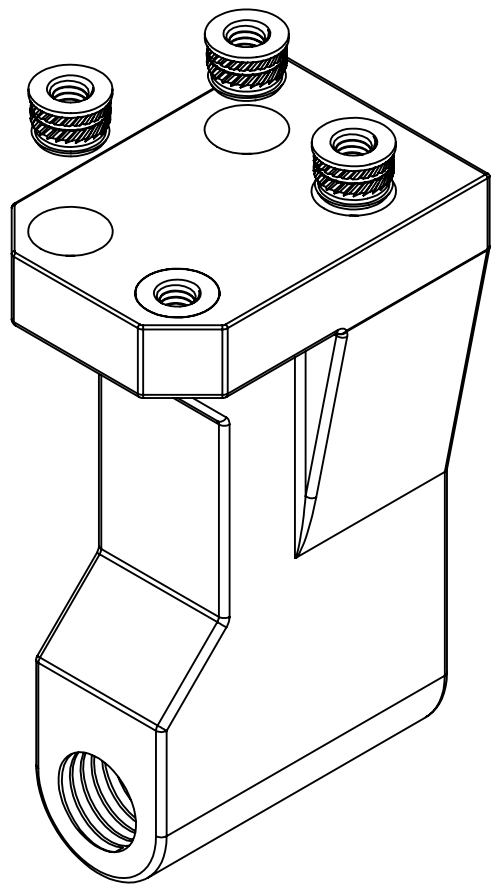
Attach each frame with the ROBOTIS M2.5 self-tapping screws.

Put together four of these assemblies as shown. There should be two with the pitch servo horn and two with the yaw servo horn



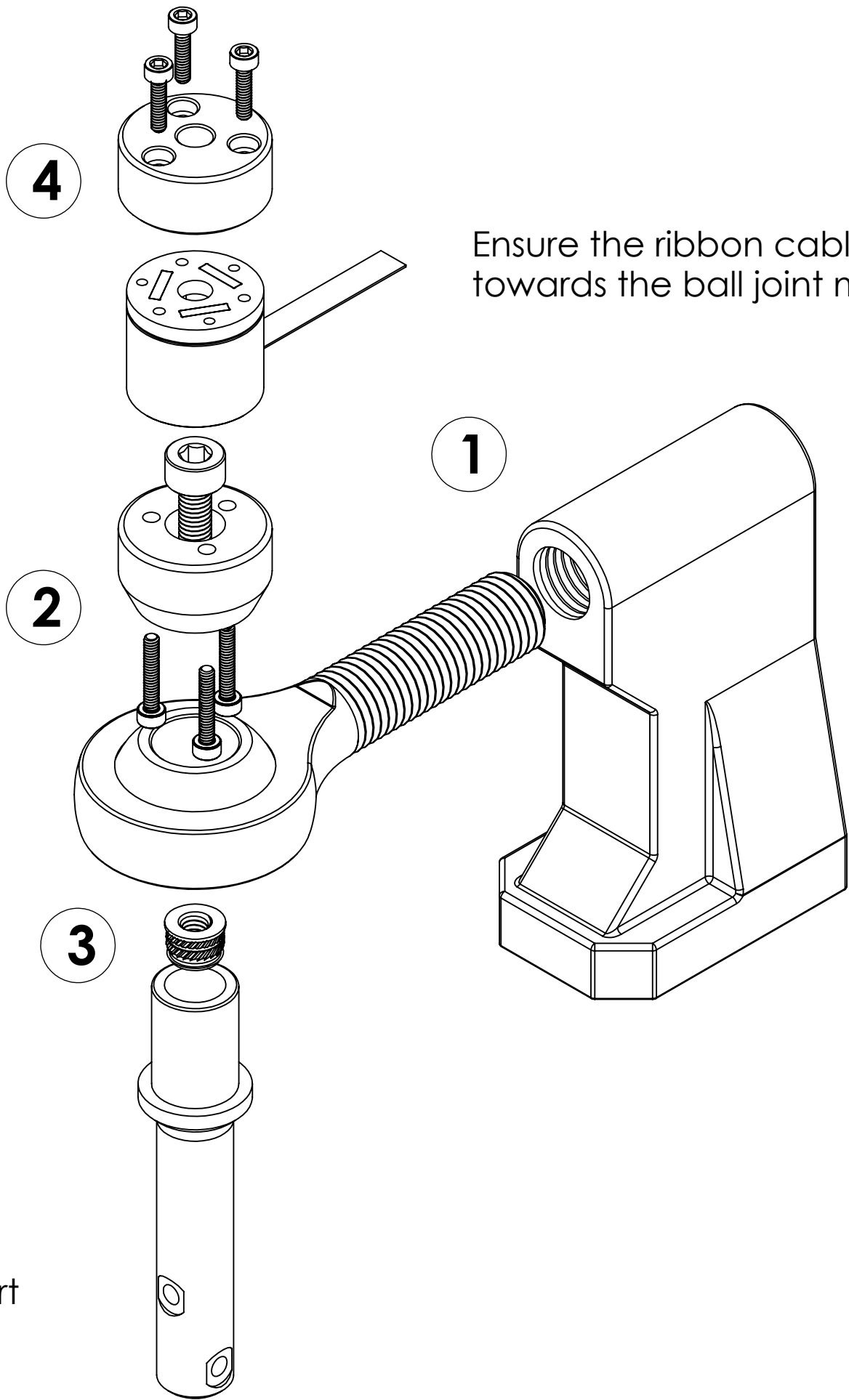
Prepare the four Dynamixel mounts with four M3 tapered heat-set inserts.

Prepare two M8 ball joint rod-end mounts with four M3 tapered heat-set inserts



Prepare the chopstick backends with one M3 tapered heat-set insert

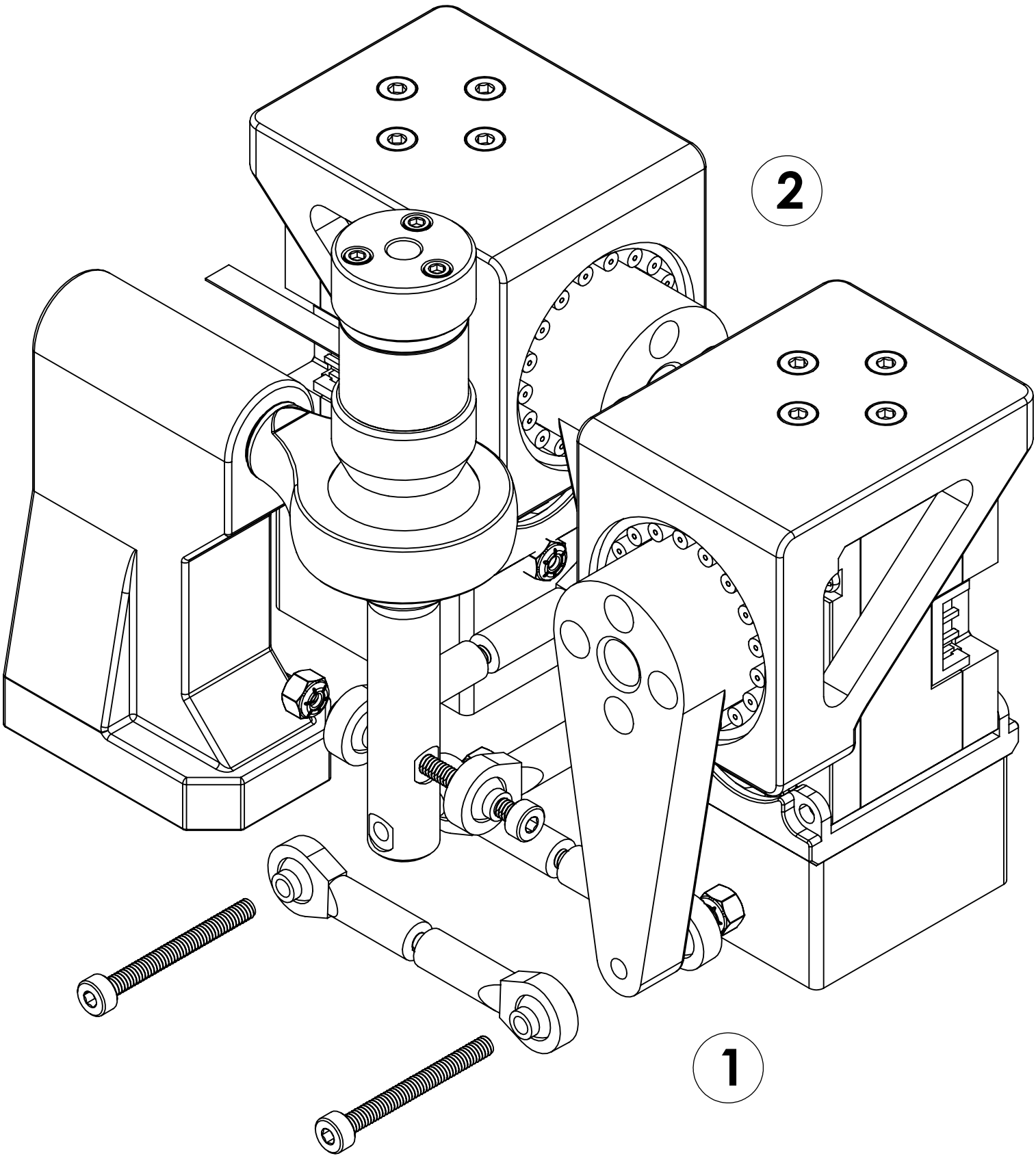
1. Insert the rod end into the mount
2. After inserting the M3 bolt into the HEX12 Mount, attach the HEX12 using three M1.4x8 bolts
3. Insert the chopstick backend into the spherical rod-end, and then by carefully inserting an allen key through the shaft of the HEX12, tighten the M3 bolt into the backend
4. Attach the 1/4" chopstick mount to the HEX12 with three M1.4x8 bolts



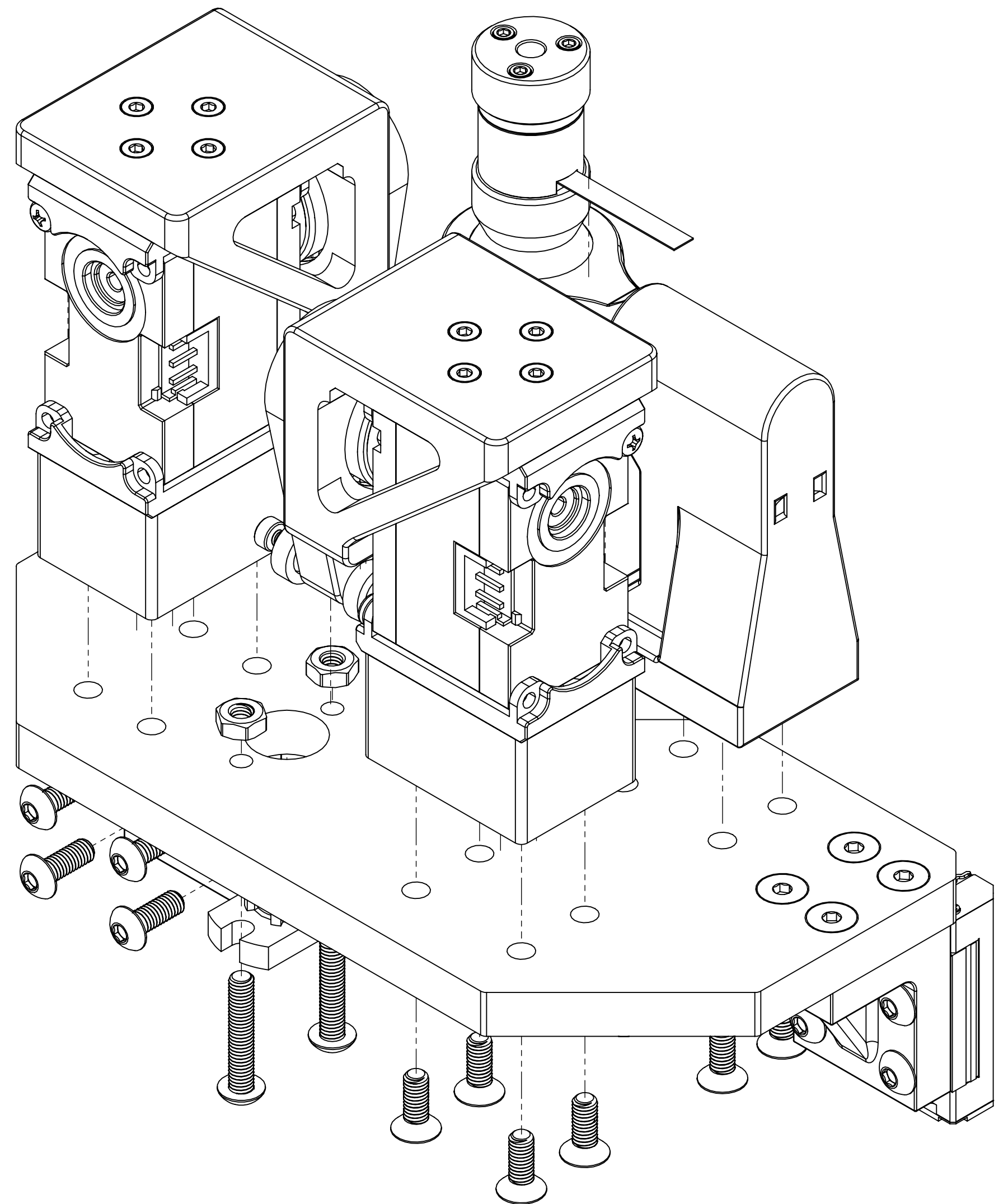
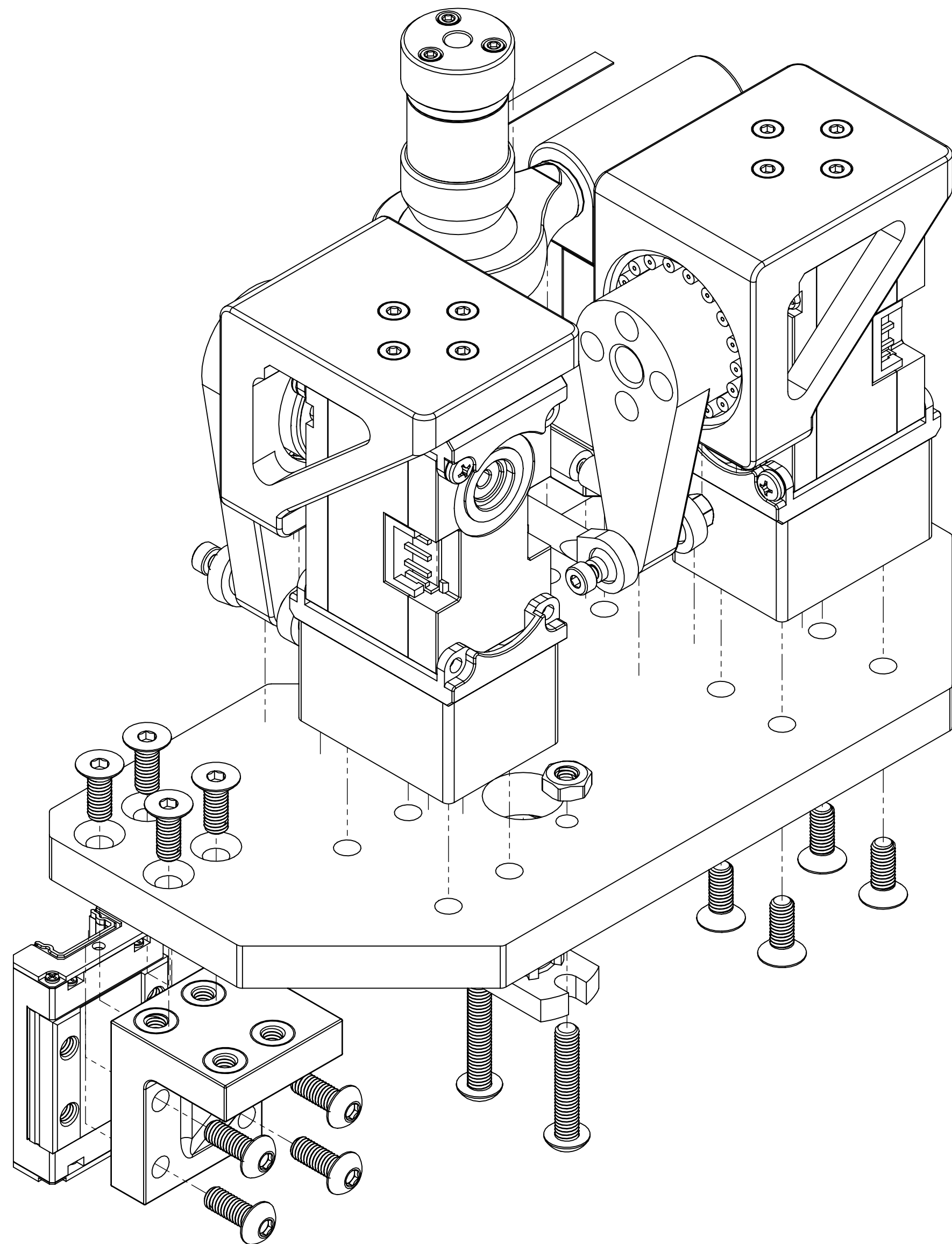
To prepare the M2 ball-joint linkages, connect two individual rod-ends with an M2x12 threaded rod
The center-to-center distance should measure 32.5mm
Check that each linkage is the correct length by using the included jig
The linkage should slot into the jig with little resistance.

1. Maintaining the assembly orientation below, connect the Yaw Servo Horn to the bottom hole in the chopstick backend with two M2 ball-joint linkages, M2x20 bolts, and M2 locknuts. The locknuts should be barely tight so there is no play between the ball-joint linkages, but not overtight
2. Repeat with the Pitch Servo Horn and the top hole in the chopstick backend
3. Repeat once more so you have two completed assemblies

add in linkage and jig views



1. Attach two each of the right and left angle-bracket carriage mounts with 4 each M3x8 button-head bolts
2. Attach two of the assemblies to each platform with four each M3x8 button-head bolts
3. Attach a leadscrew nut to each platform with two each M3x14 button-head bolts and M3 nuts
4. To each platform, attach the entire servo and chopstick subassembly with four M3x8 button-head bolts under each mounting point



Leadscrew Servo Assembly (x2)

Prepare each with 2 M3
heat-set inserts

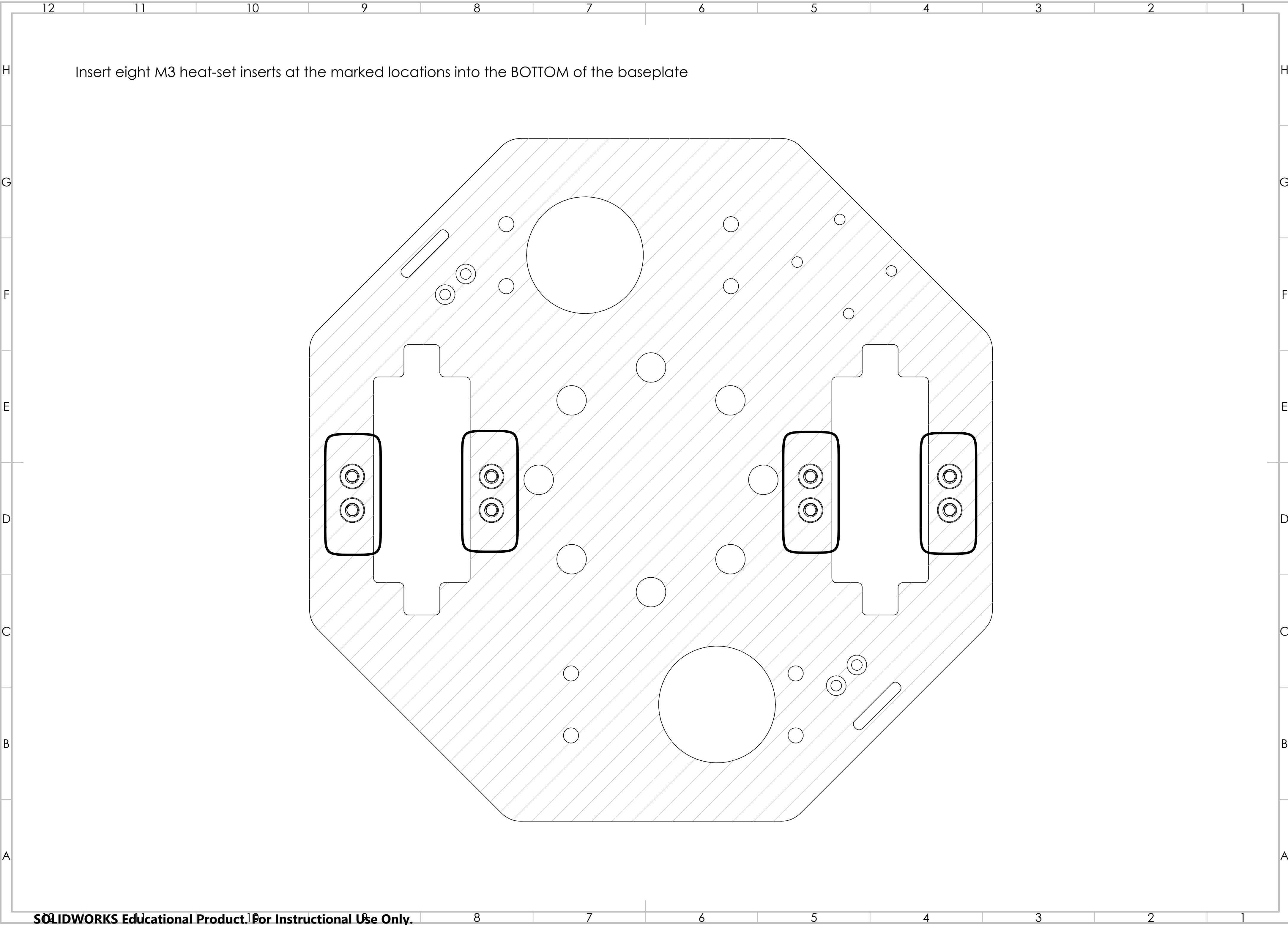
Prepare with 4 M3
heat-set inserts

9mm Rail Mount (x2)

Prepare with 8 M3
heat-set inserts

1. Attach two leadscrew servo mounts to the Dynamixel frames with four each M2x6 socket-head bolts
2. Attach the leadscrew flange coupling mount to the Dynamixel faceplate with four M2x8 socket-head bolts
3. Attach the leadscrew flange coupling to the mount with four M3x6 button-head bolts
4. Attach the leadscrew to the flange coupling by tightening the set-screw

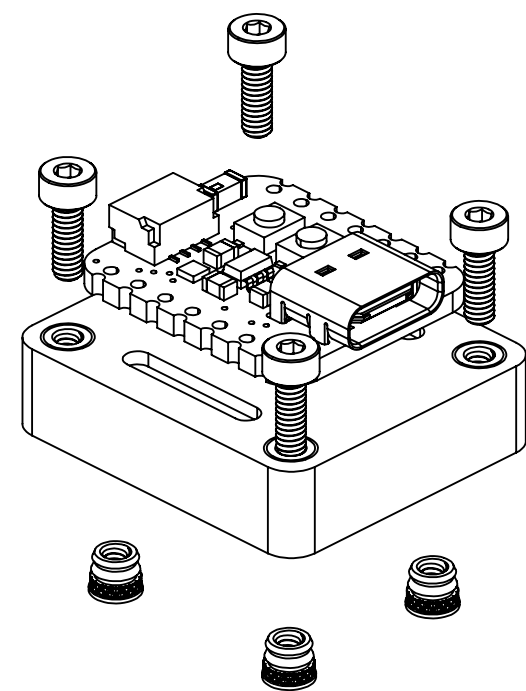
1. Using two 3D-printed alignment jigs, attach a 9mm rail to each side of the rail mount with 4 M3x8 socket-head bolts



Insert eight M3 heat-set inserts at the marked locations into the BOTTOM of the baseplate

QTPy Microcontroller Mount

Secure the QTPy to the mount with four M2x8 socket-head bolts



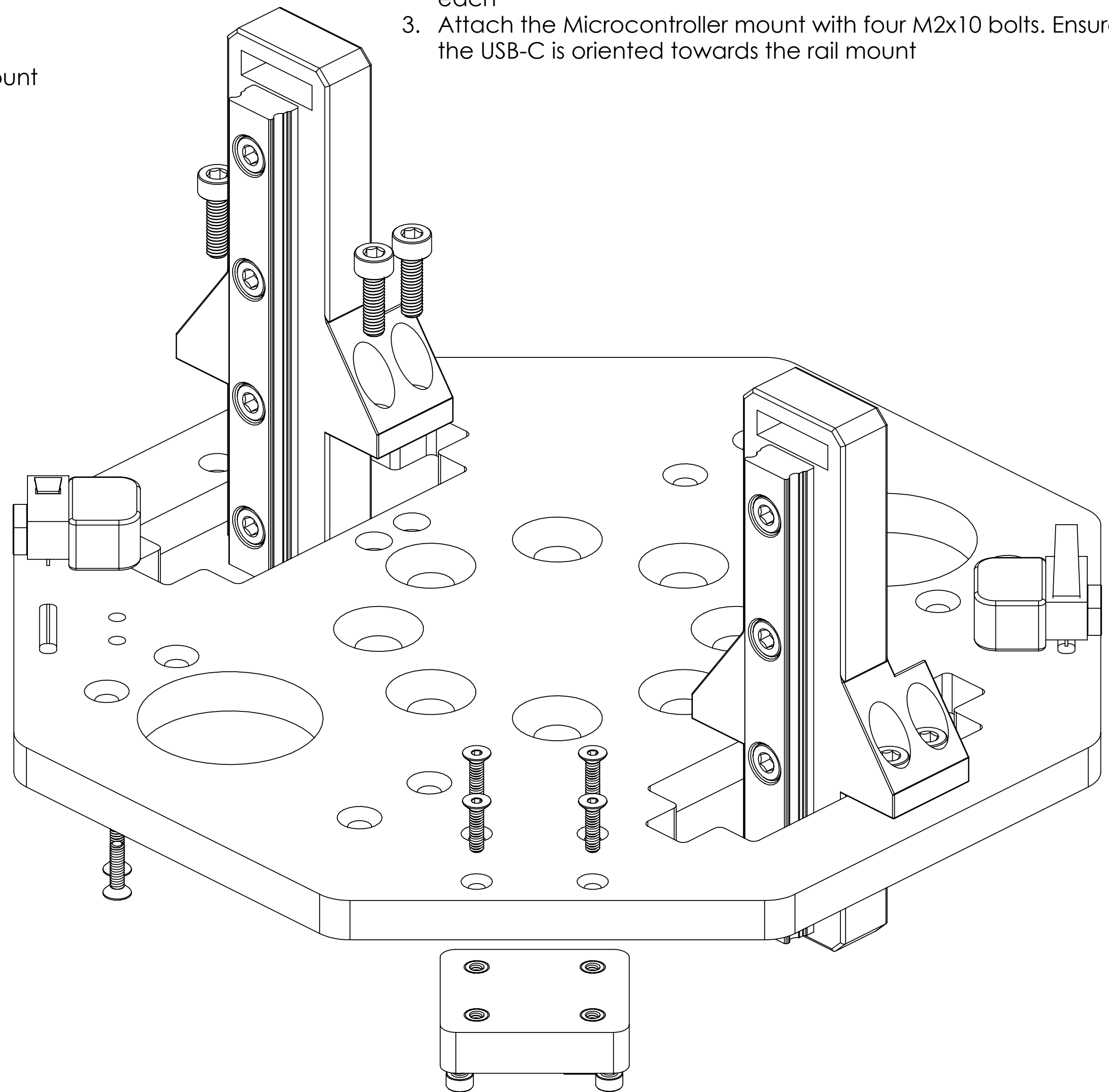
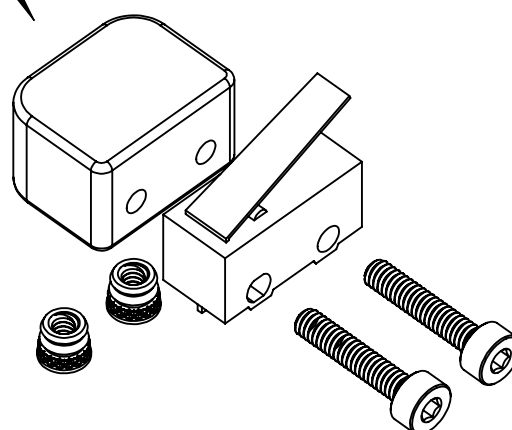
Prepare both sides of the QTPy mount with four M2 heat-set inserts each

1. Attach the two 9mm rail mounts to the platform with four M2x10 socket-head bolts
2. Attach the two Z-Endstop subassemblies with two M2x10 bolts each
3. Attach the Microcontroller mount with four M2x10 bolts. Ensure the USB-C is oriented towards the rail mount

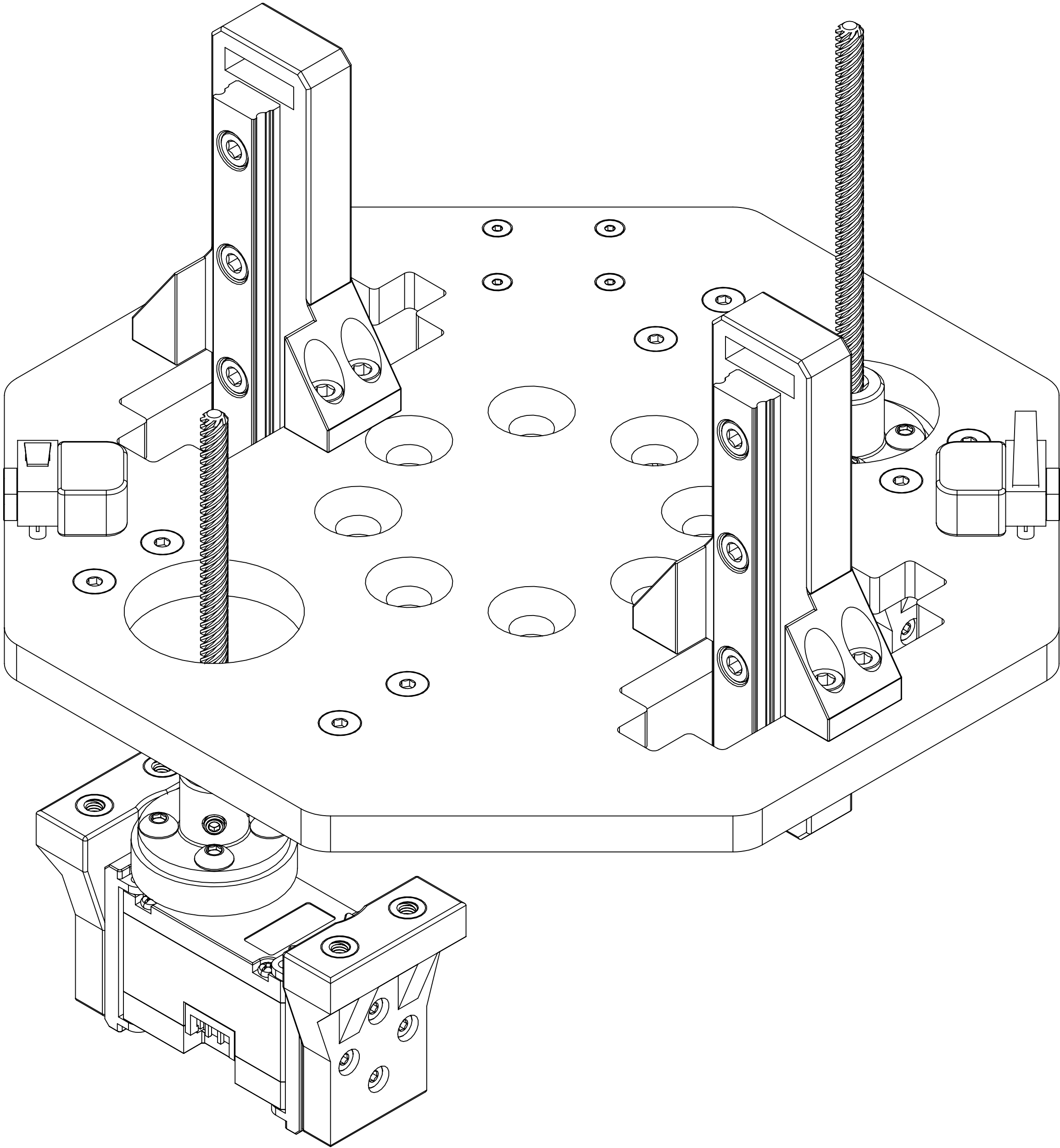
Z-Endstop Microswitch (x2)

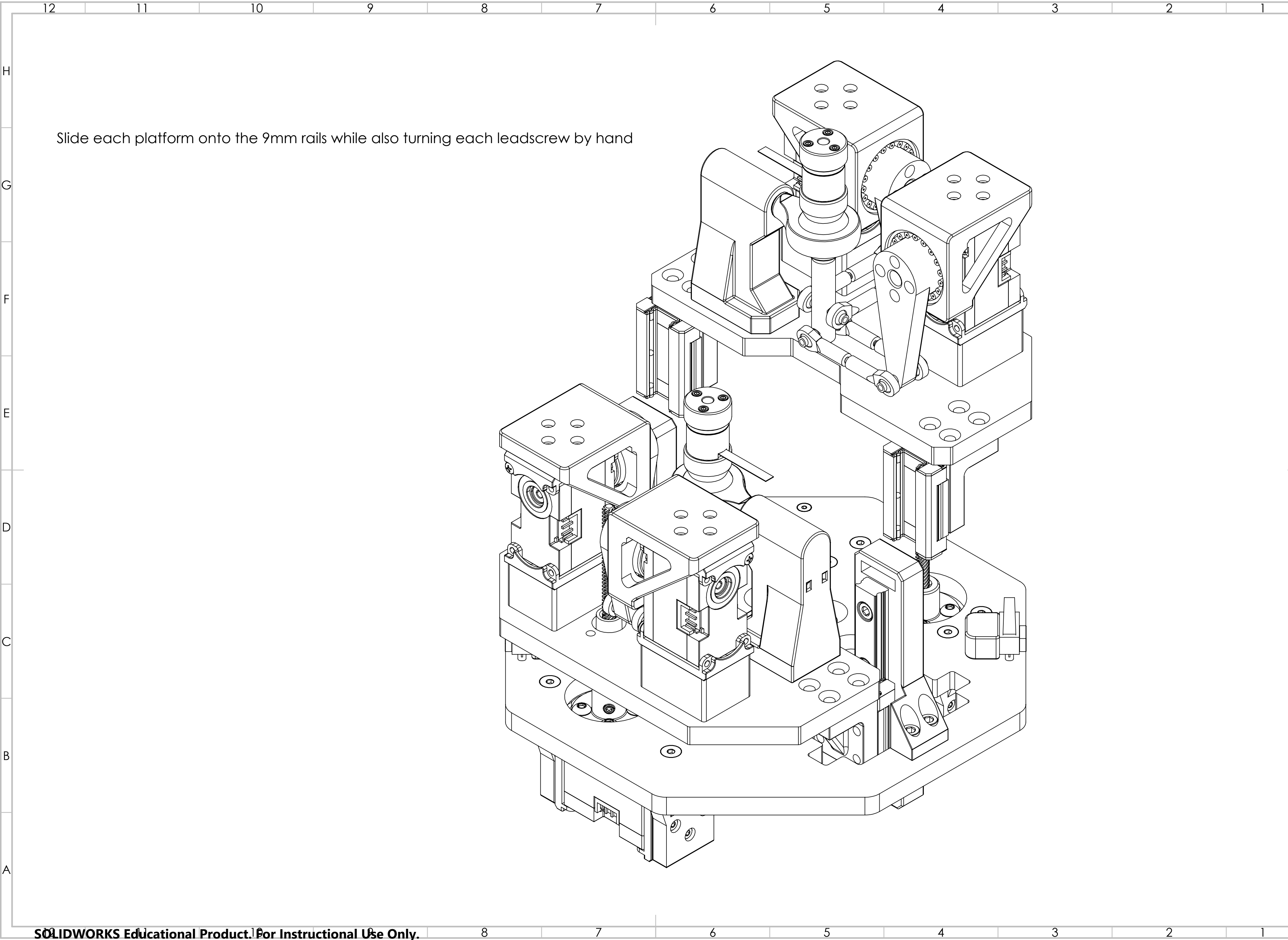
Secure the microswitch to the mount with two M2x8 bolts

Prepare the microswitch mount with two M2 heat-set inserts



Attach the two leadscrew dynamixel subassemblies to the baseplate with four M3x8 bolts each





Slide each platform onto the 9mm rails while also turning each leadscrew by hand

Screw in both chopsticks (M4 connecting rods) and you're done!

